Predicting Tomorrow's Wildfire

Comparing tree models and neural networks on real wildfire data

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Can we predict where a wildfire will burn tomorrow?

Today

Where the fire is, the weather, the land and the vegetation

→

Our models

Learn from thousands of past fires

→

Tomorrow

The chance that each 1 km square is on fire

Our data: real wildfires seen from satellites

18,545

snapshots of US wildfires, 2012 to 2020

64 km

wide area in each snapshot, split into 1 km squares

1%

of squares are burning the next day

Fire is rare, so accuracy would fool us

99%

accuracy for a model that always says “no fire”. It sounds great, but it is useless.

So we use PR-AUC

It measures how well a model finds real fires without too many false alarms. Higher is better. Random guessing scores about 0.013.

We compared three kinds of model

Tree models

Look at one square at a time, using 32 facts we designed, like distance to the fire and wind direction.

Neural networks

A CNN and a U-Net look at the whole map and learn patterns themselves.

Hybrid

Combines a tree model and a network, to see if each can help the other.

How the U-Net reads a map

1. Shrink

Compress the map to understand the big picture.

→

2. Grow back

Expand to full size, giving an answer for every square.

→

3. Keep the detail

Skip connections carry fine detail, like the fire's edge, straight across.

We tested on data the models had never seen

New weeks

The standard test. But about 95% of its areas also appear in training.

New places

Test areas never appear in training.

New weeks and places

The hardest test, but also the smallest, with only 405 snapshots.

We blocked a hidden shortcut

0.257

U-Net, trained on random crops so it cannot use position

0.341

U-Net on full images. Higher, but only because it learns “the middle burns”.

Result: tree models won all three tests

	New weeks	New places	New weeks and places
Best tree model	**0.265**	**0.141**	**0.124**
U-Net	0.257	0.130	0.101
Hybrid	0.259	0.140	0.115

Higher is better. Gaps are small. The U-Net is really behind; on the two harder tests the hybrid is level with XGBoost.

The hybrid helped the U-Net, but not enough

+0.010

better than the U-Net on new places

+0.014

better than the U-Net on new weeks and places

But

It did not beat the best tree model on any test.

Today's fire map matters most

0.239

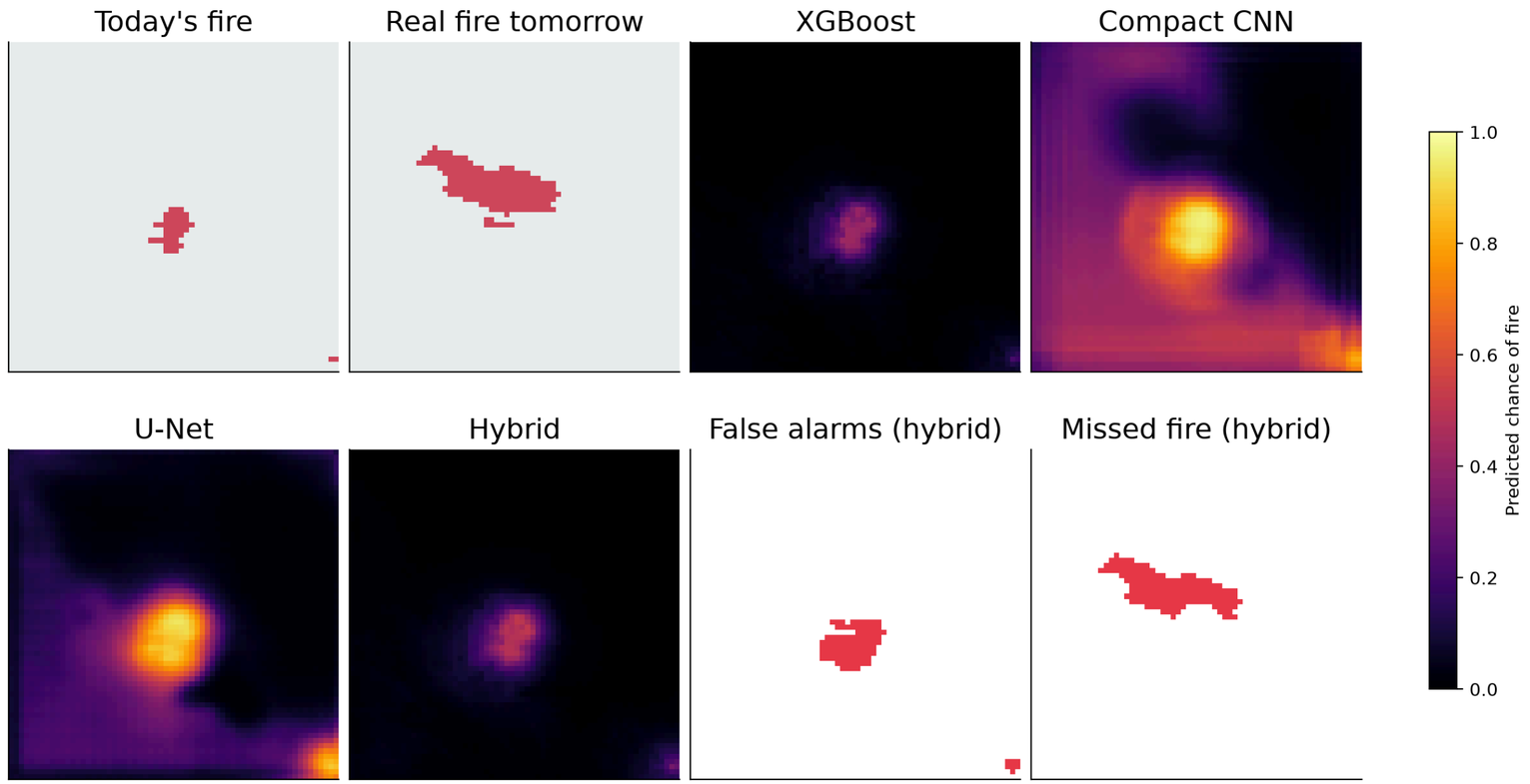
U-Net score with every input

→

0.030

after removing today's fire map

One example: the fire moved



The models expect fire to stay close to where it is today, so they miss fire that jumps further away.

What we cannot claim

- The data has no real dates, so this does not prove we can forecast future seasons
- Every snapshot already contains a fire, so this is not a search for new fires
- It shows patterns in the data, not why a fire spread

What we learned

- Well-designed features can match or beat neural networks on this problem
- How you test a model changes what you can conclude
- Combining models helped the network, but the simple tree models stayed on top